
Reorientation of a Liquid Crystalline Side-chain Polymer in Electric and Magnetic Fields Investigated by Solid-state ^1H -NMR

Peter Holstein,* Michael Bender, Mario Winkler and Dieter Geschke

Universität Leipzig, Fakultät für Physik und Geowissenschaften, Linnéstrasse 5, 04103 Leipzig, Germany

ABSTRACT

With a capacitor placed in the radio frequency coil of an nuclear magnetic resonance spectrometer it is possible to investigate reorientational processes in liquid crystalline polymers induced by the static magnetic field of the spectrometer and a dynamic electric field between the capacitor plates.

In the nematic phase, the alignment of the director is a result of the interplay between two competing fields. The director can be rotated in every desired angle by the simultaneous application of the electric field. The dynamics of the process depends strongly on the strength of the electric field and on the temperature. The arrangement allows the in situ measurement of switching times, of the rotational viscosity γ_1 , etc. © 1998 John Wiley & Sons, Ltd.

KEYWORDS: reorientation dynamics; solid-state NMR; electric fields; liquid crystalline polymers; LCP 105

INTRODUCTION

Side-chain liquid crystalline polymers have been considered as a class of substances that combine some favorable properties of low molecular weight liquid crystals and of polymers [1]. In particular, macroscopic orientations over larger areas can be used in displays. In practice, rewritable orientation is achieved by the application of electric and magnetic fields. In low molecular weight liquid crystals, relatively fast switching effects can be achieved. In contrast, caused by the macromolecular properties, liquid crystalline polymers (LCP) have much larger reorientation times in external fields. However, switching times in nematic LCP can be much shorter than seconds at higher temperatures. The investigation of fast reorientations in external fields requires modifications of nuclear magnetic resonance (NMR) techniques [2].

NMR has proved to be a powerful method to investigate orientational states [3]. The resonances of various nuclei (^1H , ^2H , ^{13}C) have been used to get information about the orientation in anisotropic materials. The advantage of the high sensitivity of the ^1H -NMR is offset by the drawback of the superposition of the strong dipolar interaction of many protons in each molecule and between adjacent ones, which provides often unresolved spectra. In contrast, deuterium resonance spectra

* Correspondence to: Dr. P. Holstein, Universität Leipzig, Fakultät für Physik und Geowissenschaften, Linnéstrasse 5, 04103 Leipzig, Germany.

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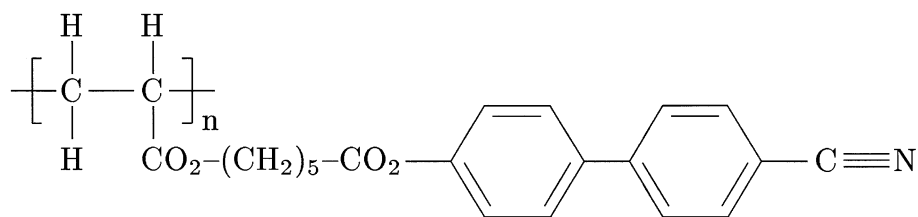


FIGURE 1. Nematic side-chain polymer LCP105 $n = 4880$, $M_w = 8030$, $n \sim 13.5$.

are extremely sensitive to orientation and molecular mobility on the deuterated sites. Recently, multidimensional ^{13}C -CP/MAS techniques (cross-polarization, magic angle spinning) [4] were developed to access the information about site-resolved molecular anisotropy in liquid crystalline polymers. The application of rotor-synchronized data acquisition collecting three-dimensional data allows orientational parameter for each resolvable carbon site to be measured, as demonstrated for a frozen smectic liquid crystalline side-group polymer [5]. It has been proved that the orientational degree for the sites in the mesogenic part of the molecule is higher than for the spacer carbons, whereas the main chain site remains disordered.

Different attempts have been made to observe field-induced reorientation processes on the molecular level by means of NMR. For reasons of sensitivity and the necessity to accumulate the NMR signal, investigations are very often limited with respect to the time scale of orientational effects. Typically, access to the dynamics in the order of some minutes is possible when proton resonance methods are employed [6–8]. Only few contributions have been published dealing with the simultaneous application of electric fields in NMR experiments to discuss alignment effects [2, 9, 10].

The reversibility of the switching process in electric and magnetic fields can be used to overcome the complications that arise from the use of thin capacitor samples and fast reorientation processes up to $100\mu\text{s}$ [2]. In this paper, we present an application of the method for the investigation of externally induced reorientation processes of a nematic side-chain polymer.

EXPERIMENTAL

First measurements on polymeric materials have been carried out using the thermotropic side-chain liquid crystalline polymer LCP 105 (Merck Ltd, GB, Poole) filled in a $100\mu\text{m}$ capacitor cell (Fig. 1). The polyacrylate LCP forms a nematic phase between 32°C and 130°C . Figure 2 shows the geometric arrangement of the sample in the NMR spectrometer and introduces the angles used in the discussion.

The laboratory-made experimental setup consists of a high voltage supply (max. 400V), a pulse generator and a special switching unit producing distortion-free high-voltage pulses. These pulses of variable length are applied to the capacitor. The

capacitor cell consists of two glass plates with conducting indium tin oxide (ITO) surfaces ($\text{In}_2\text{O}_3/\text{SnO}_2$) and polyimide spacers and is filled with the LCP as dielectric material [11]. The cell thickness is $100\mu\text{m}$. The relative large thickness of the capacitor has been chosen, among other reasons, to get a better signal-to-noise ratio, and to minimize the influence of surface effects in the LC cells which can be dominant if the cell thickness is $\lesssim 20\mu\text{m}$. The quality of the preparation (homogeneity of filling) has been tested by polarizing microscopy. The dielectric and magnetic anisotropies $\Delta\epsilon$ (~ 12) and $\Delta\chi$ ($\sim 1.4 \times 10^{-6}$) are positive [11]. Positive anisotropy leads to an alignment in the direction of the applied field. Therefore, the electric field must have a different direction from the magnetic field in order to manipulate orientation states.

The NMR experiments were carried out using a “Bruker-MSL100” spectrometer which was equipped with a ^1H -probe of extremely short ring-down time and improved background conditions (Bruker HP FF 100WB 1H sol5). The magnetic field equivalent to 100MHz proton frequency (2.3T) is a good compromise between the NMR

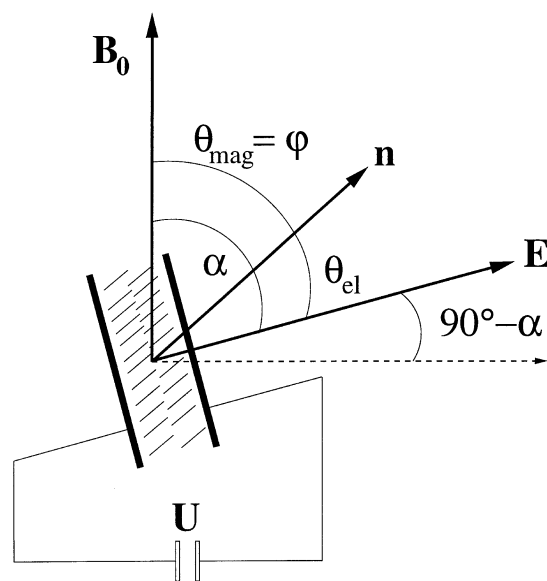


FIGURE 2. The arrangement of the capacitor relative to the magnetic field B_0 allows, by means of an applied voltage, the variation of the angle of the director n of the nematic sample. α denotes the angle between the spectrometer field and the direction of the electric field (75° in the experiment).

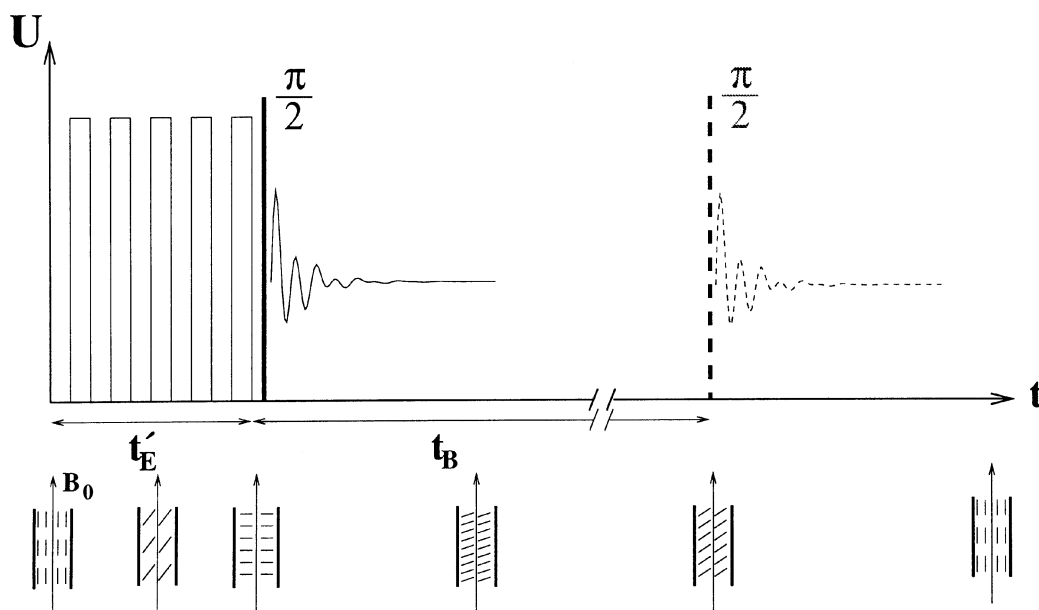


FIGURE 3. Synchronous timing of the electric field application and the start of the NMR experiment, t'_E denotes the time in which the sample is exposed to the voltage U (only $U \neq 0$ contributes to t_E , $t_E = t'_E/2$). The left 90° pulse and the following FID allows access to effects due to orientations in electric fields. The right 90° pulse and FID illustrates the option to introduce a delay (t_B) after switching off the voltage in order to measure the reorientation in the magnetic field (B_0). The 90° pulse stands symbolic for any NMR experiment (e.g. a "solid-echo" sequence as used in this investigation).

sensitivity and the orienting torque in the magnetic field B_0 . Higher magnetic fields could complicate the use of the two competing fields. The interpretation of dipolar effects becomes more complicated because of the rising influence of the chemical shift even for protons in solid-state NMR spectra. All spectra were taken with a solid-echo pulse sequence which has been synchronized with the electrical excitation. The synchronization of electric pulses with the NMR experiment is schematically sketched in Fig. 3. The variation of the delay t_B can be used for the observation of the orientation back towards the equilibrium in the magnetic field. Only the time $t_E = t'_E/2$, in which $U \neq 0$, is relevant for the orientation process in the electric field.

The application of pulsed voltages minimizes effects due to electrical currents which could be caused by a low conductivity of the material. As found experimentally by polarizing microscopy the distortions can be suppressed by the application of pulsed voltages.

During the fast reorientation in the electric field no accumulation of data is possible. The experiment can be repeated for the purpose of accumulation. The cycle *electric orientation-magnetic back orientation* can be repeated until a good signal-to-noise ratio is achieved. However, the waiting time for achieving the equilibrium state in the magnetic field B_0 can be very long. A cycle can take several minutes. It can take some hours to record one spectrum and days to get a complete reorientation curve owing to the large number of accumulations that are needed in the case of the material used. Therefore, the experimental time can be drastically shortened by recording the complete orientational process in one cycle by recording a series of FIDs

(free induction decay) without accumulation of the individual FIDs during this cycle. The FIDs have a time spacing in the order of T_1 ($T_1 = 0.45$ sec at 20°C , 0.3 sec at $100 \dots 120^\circ\text{C}$). The data accumulation becomes possible by the reversibility of the complete cycle. The reorientation process of LCP 105 in electric fields needs, depending on temperature and voltage, between 100 msec and some seconds to switch from 0° ($n \parallel B_0$) to 90° ($n \perp B_0$) when field strengths in between $0.5 \dots 4.0$ MV/m are applied across the sample.

Temperatures between 375 K and 390 K were chosen for this investigation in order to lower the rotational viscosity because we are particularly interested in the development of methods for the investigation of fast reorientation processes. For these temperatures the reorientation times range between 0.3 and 0.075 sec in the electric field. At lower temperatures the reorientation process can take essentially more time (e.g. at $T = 70^\circ\text{C}$ and 200 V in a 1.4 T magnetic field about 30 min).

RESULTS AND DISCUSSION

As an example, Fig. 4 shows the time dependence of the line shape. In Fig. 4(a), the director orients into the E -field direction; in Fig. 4(b) the reorientation goes back to B_0 .

The LCP 105 spectrum has the largest line splitting if the director is parallel to the magnetic field. The director turns if the voltage is applied. The narrow spectra appear if the dipolar interaction is minimized which is the case for the so-called "magic angle" (54.7°). A sample, in which the director would be oriented perpendicularly to the

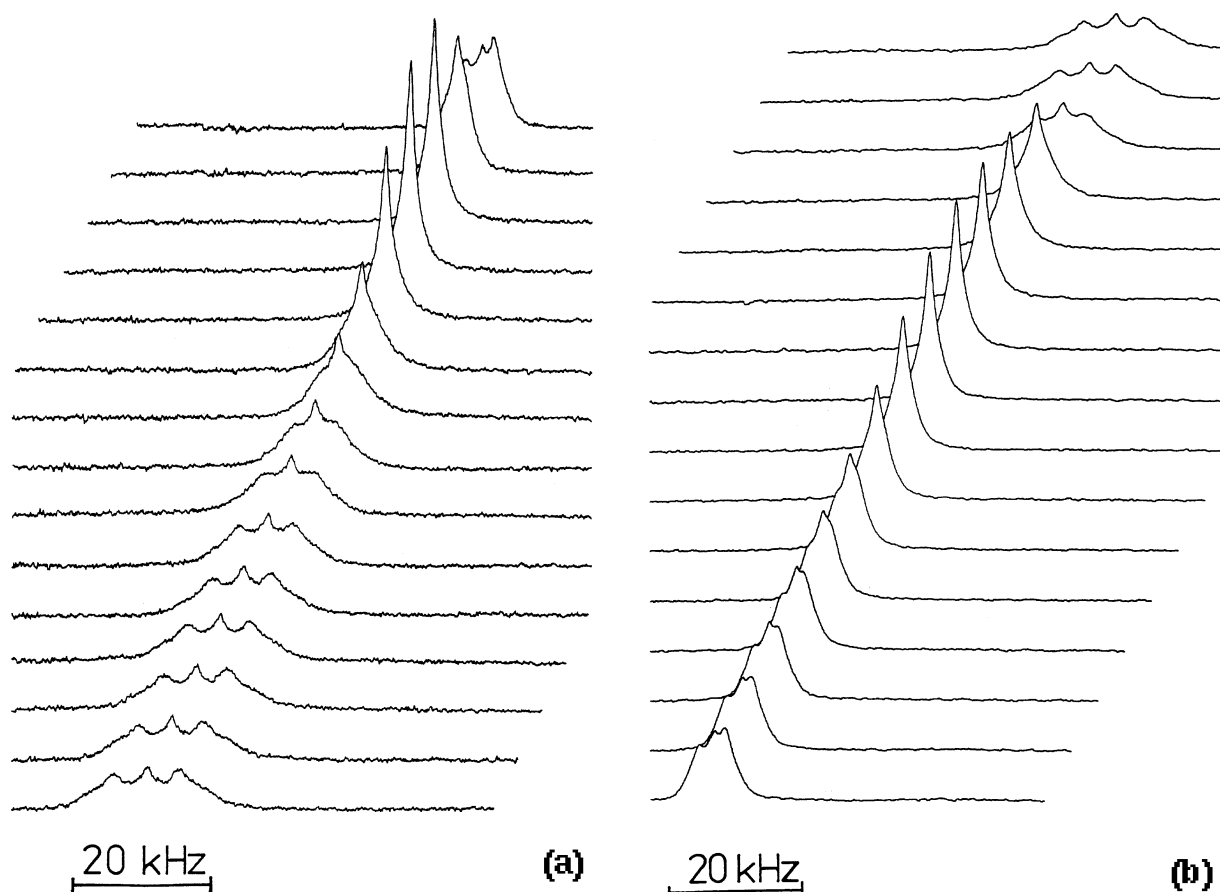


FIGURE 4. (a) Orientational behavior of a side-chain liquid crystalline polymer in an electric field (400 V across 100 μm) at 385 K. Pulses of 2 msec duration have been used followed by a 2 msec off-time. The reorientation times (E -field on, t_E in Fig. 3 are (msec, from bottom): no voltage (0), 10, 20, 30, 50, 60, 70, 80, 90, 100, 120, 140, 160, 200, 500. (b) Reorientation back to the equilibrium into the B_0 direction after switching off the E -field which has been used in (a). The time delays t_B are (sec, from bottom): 1, 1.5, 2, 2.5, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 60.

B_0 direction, had half of the splitting (eq. 1). The spectrum on top in Fig. 4(a) represents the start condition for the reorientation back to the equilibrium state (Fig. 4b). For the description of the dynamics of the proton line shapes under the influence of external fields, the following assumptions for the side-chain part of the nematic liquid crystalline polymer were made as discussed in an earlier paper [11]. Fast molecular rotations around the side-chain axis occur. The axes of the mesogenic parts of the molecules are conically distributed around the director within a nematic domain. An axially symmetric director distribution has been assumed about a preferred axis $\mathbf{n}$ in the sample (e.g. induced by the magnetic field B_0).

The NMR linewidth of proton spectra is dominated by dipolar interactions [12, 13]. The line splitting of an uniaxial liquid crystal with molecules which undergo fast rotation is given by a greatly simplified equation

$$\Delta\nu = \Delta\nu_0 \cdot \left[\frac{1}{2} (3 \cos^2 \varphi - 1) \right] \cdot S \cdot S' \quad (1)$$

with

$$\Delta\nu_0 = \frac{3\mu_0\gamma^2\hbar}{8\pi\langle r_{ij}^3 \rangle} \quad (2)$$

$\Delta\nu_0$ describes the line splitting of the sample at $\varphi = 0$ which is the case for the magnetically oriented LCP and can be measured. μ_0 is the susceptibility of the vacuum, γ is the gyromagnetic ratio of protons, $\hbar$ is Planck's constant divided by 2π , S' is the degree of orientation, S is the order parameter, r_{ij} is the distance between adjacent resonating nuclei and φ is the angle between the director $\mathbf{n}$ of the nematic phase and B_0 . To a good approximation it can be assumed that the orientational parameter S' is of order 1.

The main chain is only weakly oriented and contributes a featureless broadening of the line shape, corresponding to the number of protons and their mobility ($S \lesssim 0.03$ at 110°C). The degree of order decreases for the protons of the spacer from the protons attached to the mesogenic part to the sites attached to the main chain. It is assumed that all protons of the phenyl ring have the same degree of order providing the pronounced doublet splitting in the spectra [11].

The chemical shift is in the order of 4 ppm and

causes the weak asymmetry of the spectra. This effect can be neglected for the analysis of the line splitting. Homogeneous reorientation provided, the angle $\varphi(t)$ between the magnetic field B_0 and the director n of the sample, can be calculated from the spectral splitting

$$\varphi(t) = \arccos \left(\sqrt{\frac{2\Delta\nu(t)}{3\Delta\nu_0} + \frac{1}{3}} \right) \quad (3)$$

The comparison with theoretical models gives a time constant τ_{re} for the reorientational process in an electric or magnetic field. The Leslie equation for homogeneous reorientation processes in static fields was used [14]:

$$\gamma_1 \frac{d\theta}{dt} = \sum_i T_i \quad (4)$$

T_i are the external torques, γ_1 is the rotational viscosity and θ is the angle between T_i and the acting field. $\theta_{mag} = \varphi$ for the reorientation experiments in the magnetic field.

The normal of the capacitor (E -field direction) was placed in an angle of 75° to B_0 . Therefore, $\theta_{el} = 75^\circ - \varphi$ during the orientation in the electric field.

An angle smaller than 90° has been chosen to minimize possible flow-back effects [15, 16].

The torque in the magnetic field is

$$T_m = \frac{\mu_0}{2} \Delta\chi H^2 \sin(2\theta) \quad (5)$$

From the solution of the kinetic equation

$$\gamma_1 \frac{d\theta}{dt} = \frac{\mu_0}{2} \Delta\chi H^2 \sin(2\theta) \quad (6)$$

the time constant

$$\tau_{re} = \frac{\gamma_1}{\mu_0 \Delta\chi H^2} \quad (7)$$

may be obtained.

Analogously, the electric field approach with the torque T_{el}

$$T_{el} = \frac{\varepsilon_0}{2} \Delta\varepsilon E^2 \sin(2\theta) \quad (8)$$

and the kinetic equation

$$\gamma_1 \frac{d\theta}{dt} = \frac{\varepsilon_0}{2} \Delta\varepsilon E^2 \sin(2\theta) \quad (9)$$

gives the time constant

$$\tau_{re} = \frac{\gamma_1}{\varepsilon_0 \Delta\varepsilon E^2} \quad (10)$$

Experimentally, the τ_{re} were obtained by fitting the measured values to the solution of the Leslie equation

$$\varphi(t) = \arctan \left[(\tan \varphi_0) \exp \left(-\frac{t}{\tau_{re}} \right) \right] \quad (11)$$

Then, the rotational viscosity γ_1 can be calculated if $\Delta\chi$ and $\Delta\varepsilon$ are known.

A comparison of the torques in the experiments

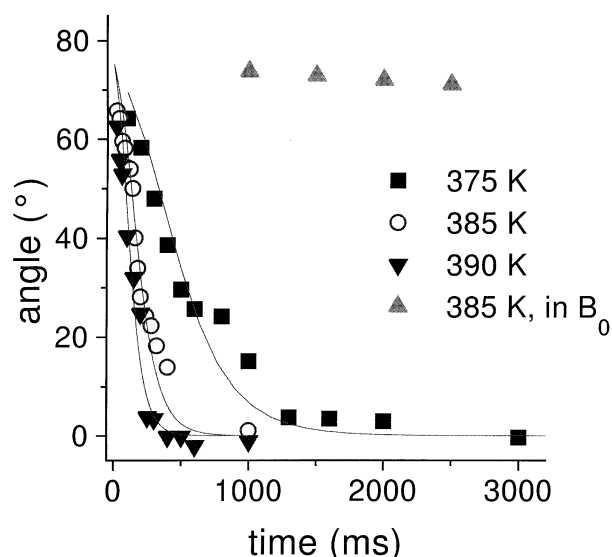


FIGURE 5. Reorientational behavior of the angle of the director in the nematic LCP 105 sample relatively to the electric or the magnetic field in dependence on the time with the temperature as parameter. The amplitude of the electric field is 400 V for all curves. For comparison the back reorientation into the B_0 -direction is given.

yields, using eqs (5) and (8) with $E = 4 \text{ MV/m}$ and $B = 2.3 \text{ T}$, a ratio of $T_{el}^{\max}/T_{mag}^{\max} = 265$. Therefore, the influence of the magnetic field can be neglected for the discussion of the reorientation process in the electric case. The magnetic field provides only defined start conditions (director parallel to B_0). The electric and the magnetic torques become comparable only at angles (n, E) less than $\sim 0.1^\circ$ (tilt of the E -field to the director of the nematic LCP), so an completely parallel (E -field) orientation can be assumed. The main splitting of the doublet has been used to calculate the angle of the director of LCP 105 relative to E or B_0 using eq. (3). The time behavior of the angle $\varphi(t)$ (Fig. 5) has been fitted with eq. (11).

It should be pointed out that there is a discrepancy (see Table 1) between the viscosities obtained from the orientations in the electric field and back towards B_0 . One reason could be the use of the $\Delta\chi$ and $\Delta\varepsilon$ values at higher temperatures. Both parameters have been scaled with the order parameter [17] in the same way, which may not be correct. Owing to the multiple particle character of the dipole-dipole interaction in abundant spin systems, the possibilities are relatively limited to discuss a differentiation in the orientational behavior of distinct parts of molecules by means of ^1H -NMR alone. In contrast to low molecular weight systems, the degree of order changes with the position in the molecule (e.g. for positional order in a side chain) which could be nicely demonstrated by multidimensional solid-state NMR [5].

Obviously, and possibly caused by the small degree of polymerization ($n \sim 13.5$), the main chains of LCP 105 take part in the orientational process but at a lower extent. The data are summarized in Table 1. The reorientation processes could be described

TABLE 1. Switching Times τ_{re} and viscosities γ_1 obtained at Different Temperatures

Temperature (K)	$S(T)/S(20^\circ\text{C})$	τ_{re} (msec)	γ_1 (Pasec)
375 (in E)	0.5	277 ± 11	235 ± 13
385 (in E)	0.43	112 ± 5	82 ± 4
385 (in B_0)	0.43	$(8.3 \pm 0.3) \times 10^3$	23 ± 1
390 (in E)	0.4	75 ± 5	51 ± 5

E and B_0 denote the orientation in the electric and magnetic field, respectively. γ_1 has been calculated using eqs (7) and (10) with $\Delta\chi(T) = (S(T)/S(20^\circ\text{C})) \times 1.4 \times 10^{-6}$ and $\Delta\varepsilon(T) = (S(T)/S(20^\circ\text{C})) \times 12$; $B = 2.3\text{ T}$, $E = 4\text{ MV/m}$.

by a single parameter for the orientation in the electric field and in the magnetic field as well which proposes a homogeneous orientation process.

CONCLUSION

It was possible to observe reorientational processes in an electric field induced by switching of voltage across a capacitor sample. This could be achieved by the synchronization of an electric field application and the NMR pulse programs in combination with special data acquisition techniques and the reversibility of the reorientational processes. The techniques allowed the investigation even of fast switching times in the order of a hundred milliseconds. The electric field approach has been used to originate defined orientational states of the director for reorientation experiments in magnetic fields. The reorientation in magnetic fields of 2.3 T is much slower (some seconds to several minutes). A number of measurements in the range of 375–390 K has been carried out to investigate the temperature dependence of switching times. From the spectra the angle between the director and the magnetic field can be calculated. The time dependence of the angles (n, B_0) and (n, E) in the nematic liquid crystalline polymer can be described with the Leslie equation. From the data, rotational viscosities were obtained. A discrepancy in the viscosity data between the electric and the magnetic case could not be explained on the basis of the model. Further investigations are in progress.

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